

TRY IT: MAKE MOTION VISIBLE



PREDICT → OBSERVE → EXPLAIN

Prediction: _____

Observation: _____

1

MARK

Align the rotating head with the zero mark. Keep hands clear.

2

PUSH

Apply a gentle sideways push for one second. Watch the direction.

3

RELEASE

Let go. Observe whether motion continues, slows, or stops.

4

COMPARE

Repeat with a stronger push, then push the opposite way.

5

EXPLAIN

Longer arrow = larger force. A changing motion means net force or torque.

SAFETY

Use only the designated handle. Do not force, lift, unplug, or adjust the lenses.

Question to carry forward: What changed — speed, direction, or both?

CAN YOU READ THE FORCES?

1

Two equal arrows point in opposite directions. What happens to the motion?

A It must speed up

B No change in velocity

C It must turn clockwise

Lift the flap / check after you answer

2

The orange push arrow becomes longer while friction stays the same. What should you observe?

A Greater acceleration

B Less mass

C Gravity disappears

Lift the flap / check after you answer

3

You release the handle and the head slows. Which force best explains the change?

A Inertia

B Friction or resistance

C Balanced gravity

Lift the flap / check after you answer

ANSWER KEY: 1 — B • 2 — A • 3 — B